FINAL PROJECT REVIEW | BAMAT207

Breast Cancer Classification using Probabilistic Machine Learning

From initial probability models to statistical testing, model tuning and explainable results

569

Tumour samples; 30 numerical features

98.25%

Highest held-out test accuracy (Logistic Regression)

p = 0.0312

Exact planned McNemar comparison: LR versus GNB

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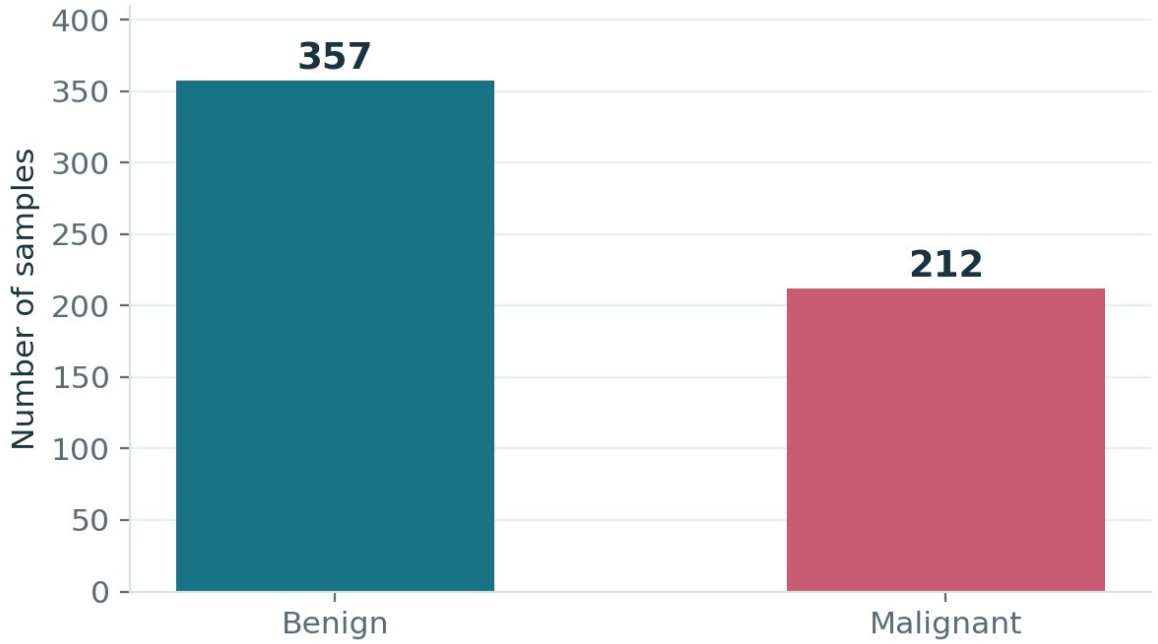
What exactly are we predicting?

Each row is a pre-extracted numerical description of cell nuclei - not an input medical scan.

PROJECT OBJECTIVE

Given 30 measurements from a biopsy sample, predict whether the tumour is malignant or benign, along with a probability.

569 samples	**30** features
212 malignant	**357** benign



A model that always predicts “benign” would already reach 62.7% accuracy; therefore we also evaluate recall, errors and ROC-AUC.

Dataset: Wisconsin Diagnostic Breast Cancer (WDBC), via scikit-learn / UCI; no missing values.

Two probabilistic models, one fair experiment

We applied the same train-test split to both models, and kept scaling inside training and CV.

Logistic Regression (LR)

Learns a weighted score from 30 standardized inputs. The sigmoid turns that score into P(benign). Coefficients are fitted by penalised maximum likelihood.

$$p = 1 / (1 + e^{-z}) \quad | \quad z = \beta_0 + \Sigma\, \beta_j x_j$$

Gaussian Naive Bayes (GNB)

Starts from class priors and combines Gaussian feature likelihoods using Bayes' theorem. It assumes the features are independent once the class is known.

$$P(\text{class} \mid x) \propto P(\text{class}) \times \prod P(x_j \mid \text{class})$$

EXPERIMENTAL PIPELINE

1 Load and explore

569 × 30 table; plots and correlation

2 Stratified split

455 train / 114 test (80:20)

3 Scale safely

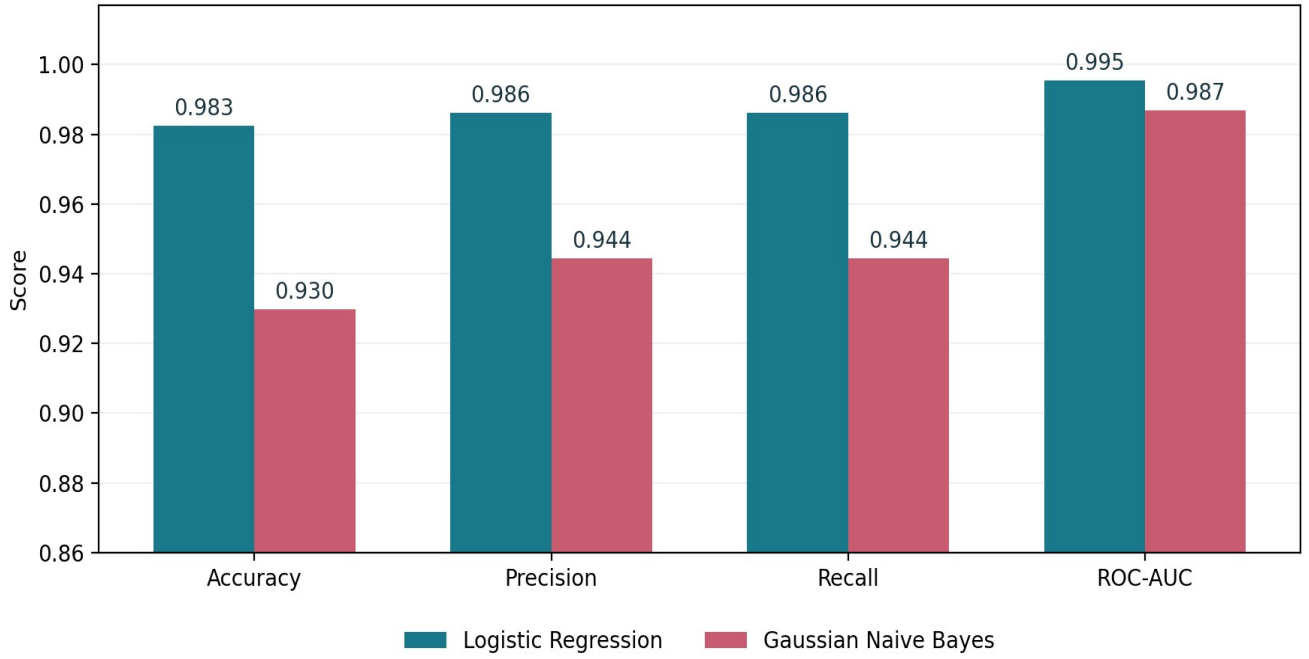
Fit StandardScaler on training only

4 Train + evaluate

Confusion matrix, AUC and 5-fold CV

Review 1 results: Logistic Regression outperformed Naive Bayes

We evaluated both classifiers on the same 114 unseen samples. Precision and recall here use benign as the positive class.



98.25% vs 92.98%

Accuracy on the 114-sample test set (LR vs GNB)

1 vs 4

Malignant tumours wrongly predicted as benign, out of 42

WHY NAIVE BAYES PERFORMED WORSE

Radius, perimeter and area are strongly correlated ($r \approx 0.99$). This challenges Naive Bayes' assumption that the features provide independent evidence.

Review 1 conclusion: Logistic Regression achieved higher accuracy (112/114 correct) and missed fewer malignant cases (1 vs 4). The result applies to this dataset and test split.

Work completed after Review 1

After comparing Logistic Regression and Gaussian Naive Bayes, we tested whether the accuracy gap was meaningful, tuned the model, added Random Forest and evaluated probability reliability.

01

Tested the accuracy difference

We applied McNemar's paired test to the same test patients, estimated a 95% bootstrap confidence interval, and cross-checked the result with repeated cross-validation.

02

Tuned Logistic Regression

We compared 13 regularisation strengths (C) using 5-fold cross-validation and selected a decision threshold using F2, which prioritises malignant recall.

03

Trained a third classifier

We trained Random Forest as a non-linear alternative and used SHAP values to identify which cell measurements influenced its predictions.

04

Evaluated probability reliability

We plotted calibration and ROC curves, compared Brier scores and log-loss, and examined whether each model's predicted probabilities were dependable.

The following slides explain what we tested, what the numbers mean, and which changes did or did not improve performance.

Was the 5.26-point accuracy gap statistically meaningful?

We used McNemar's test because both models predicted the same 114 patients. The test focuses on cases where only one model was correct.

PAIRED RESULTS

	GNB correct	GNB wrong
LR correct	106	6
LR wrong	0	2

LR was correct in 6 cases GNB missed; GNB had no cases where it alone was correct.

0.0312

Exact McNemar p-value. Below 0.05 for the planned LR–GNB comparison.

+5.26 pts

LR exceeded GNB by 5.26 percentage points on the test set.

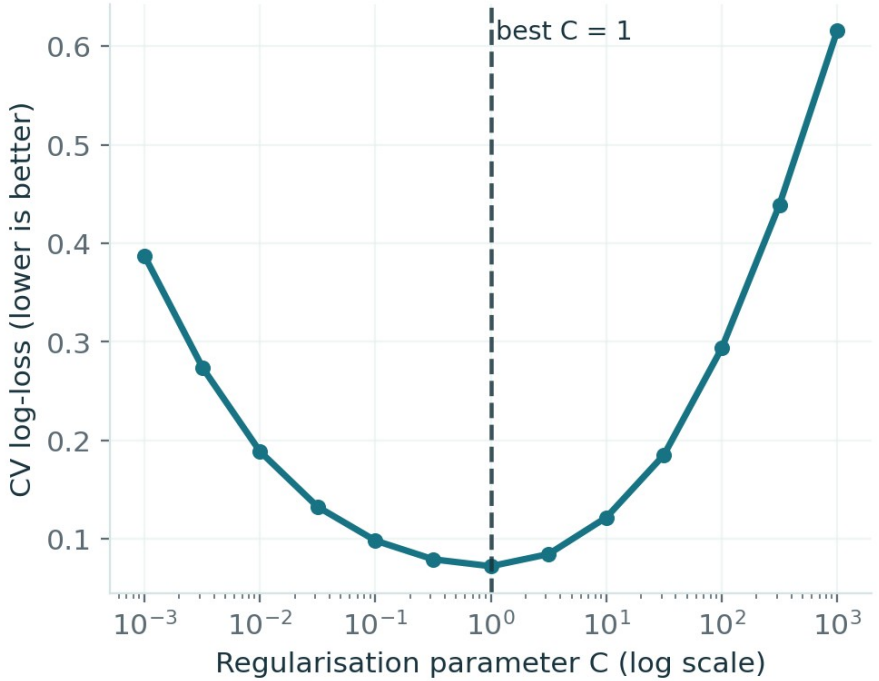
TWO SUPPORTING CHECKS

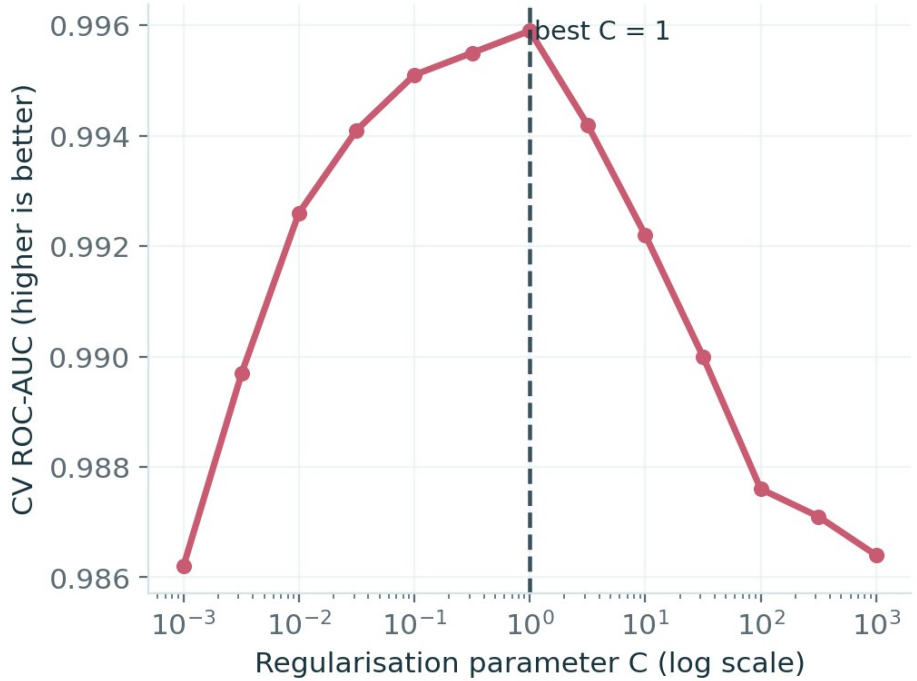
Bootstrap 95% CI: +1.75 to +9.65 percentage points
Repeated 5-fold CV: corrected t-test p = 0.0012

Important caution: across all 3 exploratory pairwise model comparisons, the Bonferroni-adjusted LR–GNB p-value is 0.0938. The unadjusted 0.0312 applies to the preplanned comparison.

Regularisation tuning: the default C was already best

We tested 13 values of C (0.001–1000) using 5-fold cross-validation on training data. Lower log-loss means more reliable probability predictions.





Best C = 1.0

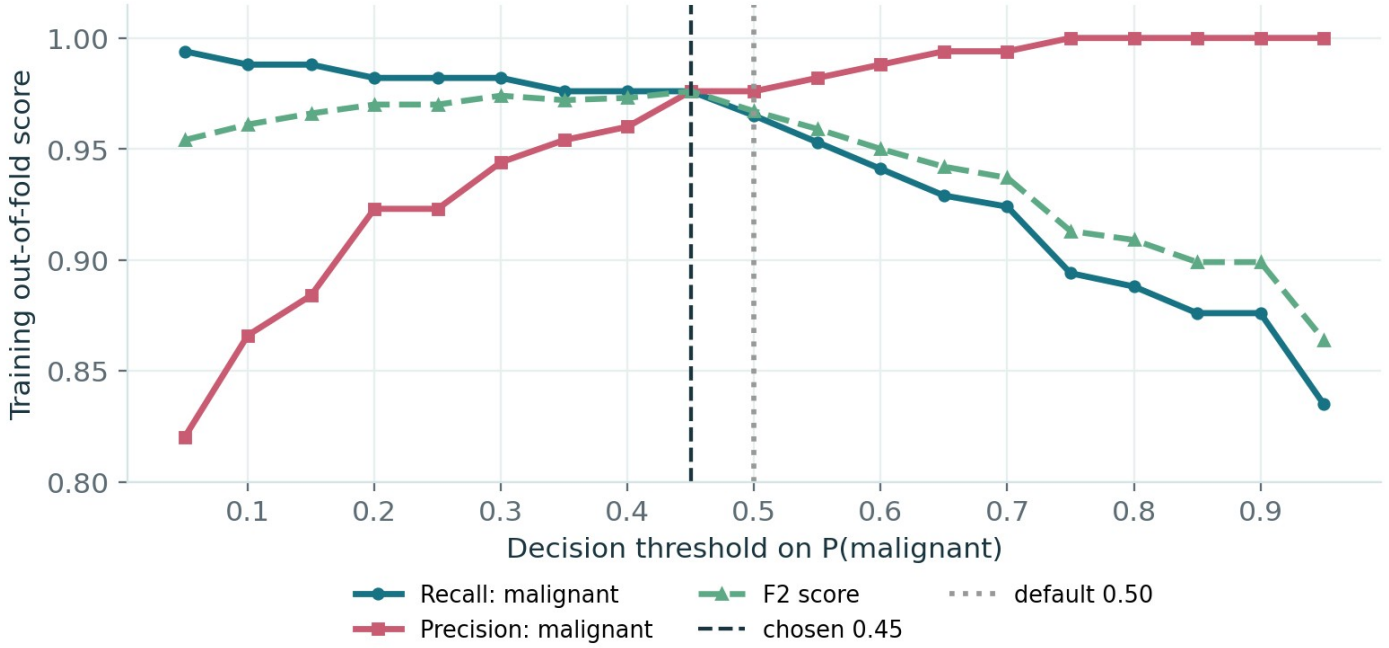
Min. log-loss = 0.0723

CV AUC = 0.9959

What we found: C = 1.0 gave the lowest validation log-loss (0.0723) and highest AUC in our search (0.9959). We retained the default instead of changing a setting that already worked.

Threshold tuning: lowering the cutoff increased false alarms

We chose threshold 0.45 by maximising F2 on out-of-fold training predictions. F2 assigns more importance to detecting malignant cases than avoiding false alarms.



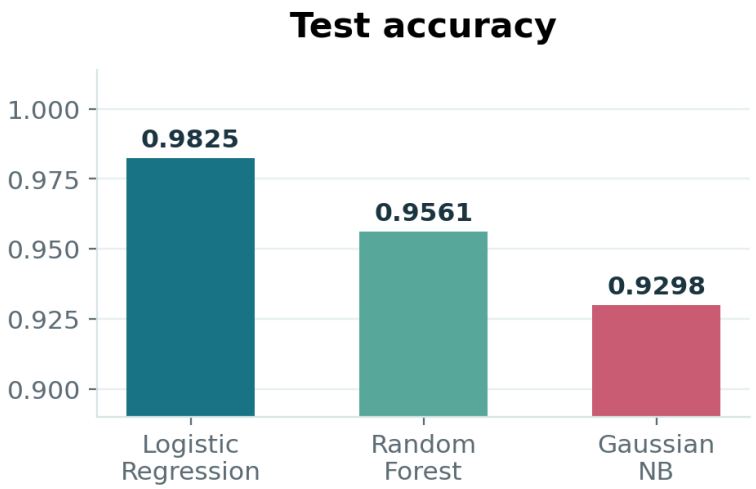
TEST-SET EFFECT

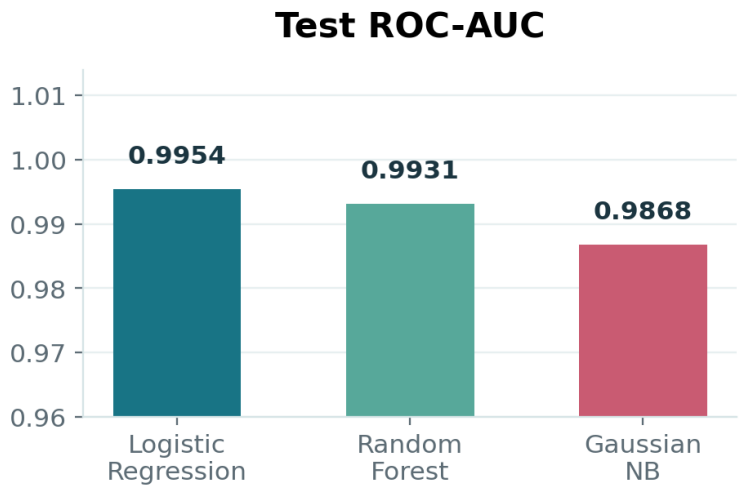
	Threshold 0.50	Threshold 0.45
Malignant caught	41 / 42	41 / 42
Malignant missed	1	1
Benign falsely flagged	1	4
Overall accuracy	98.25%	95.61%

On the unseen test set, both thresholds detected 41 of 42 malignant cases. The 0.45 cutoff flagged 3 additional benign cases incorrectly, so we retained 0.50.

Random Forest: a third, non-linear comparison model

We trained a Random Forest that combines many decision trees. Unlike the other models, it can capture non-linear interactions without assuming independent features.





95.61%

Random Forest test accuracy (109 / 114 correct)

0.9931

Random Forest test ROC-AUC

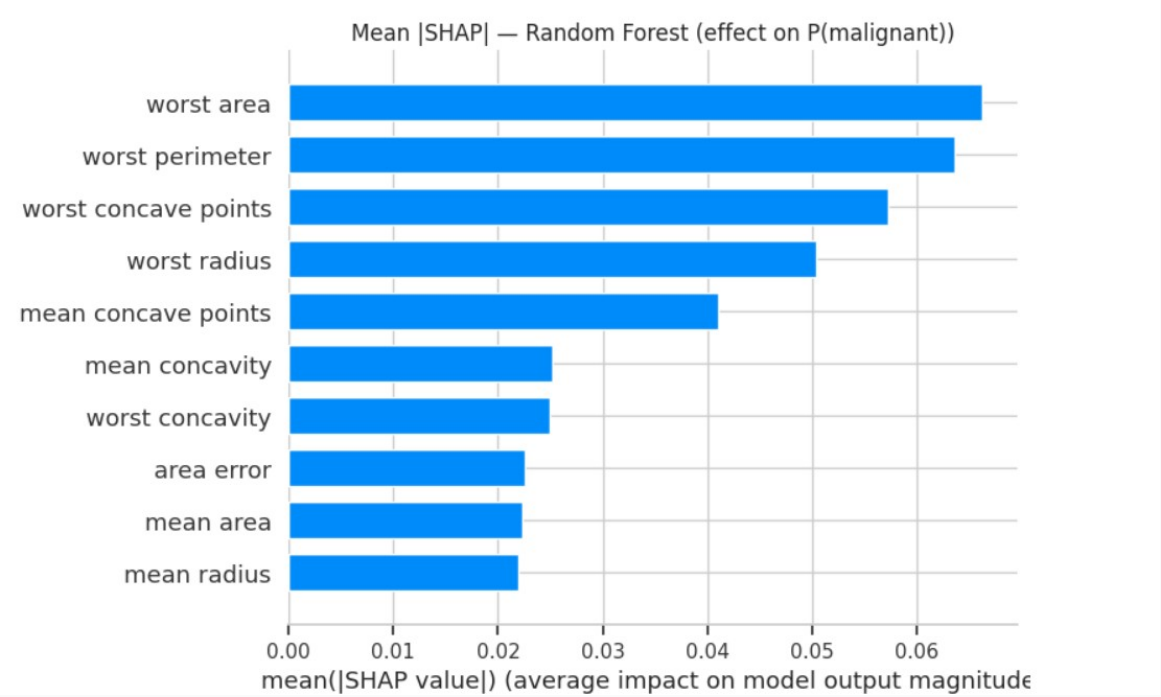
3 of 42

Malignant cases missed by Random Forest

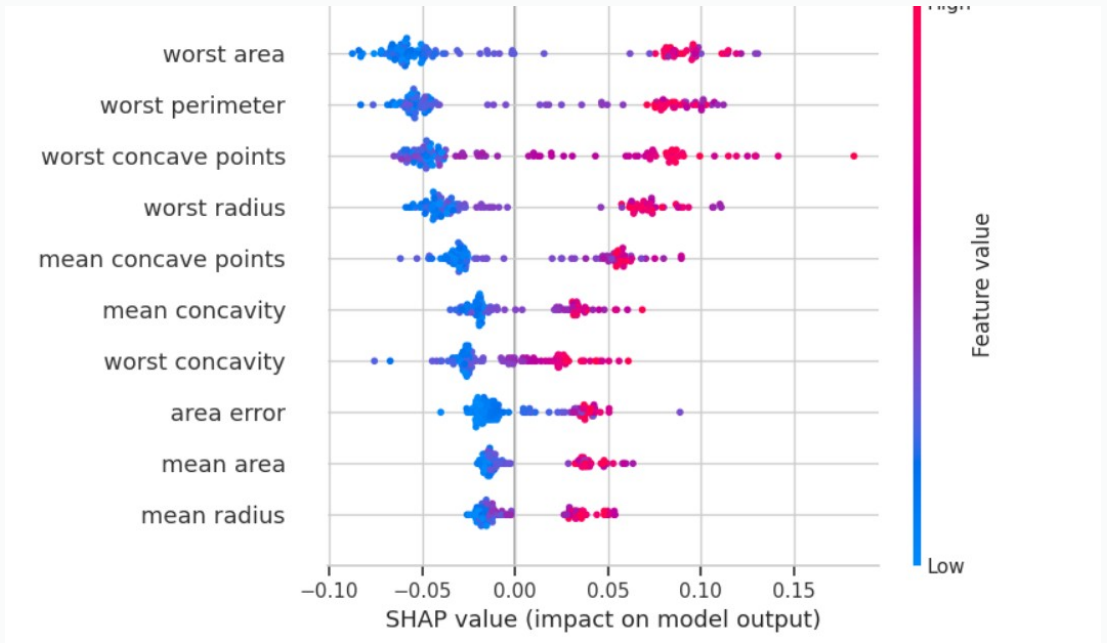
Random Forest achieved 95.61% accuracy and 0.9931 AUC on the test set, but missed 3 malignant cases. Logistic Regression still led with 98.25% accuracy and only 1 missed malignancy.

SHAP explained the Random Forest predictions

We used SHAP to estimate how much each measurement shifted a prediction toward malignant or benign. The bars show average influence; the dot plot shows the direction and variation.



Average importance (bar plot)

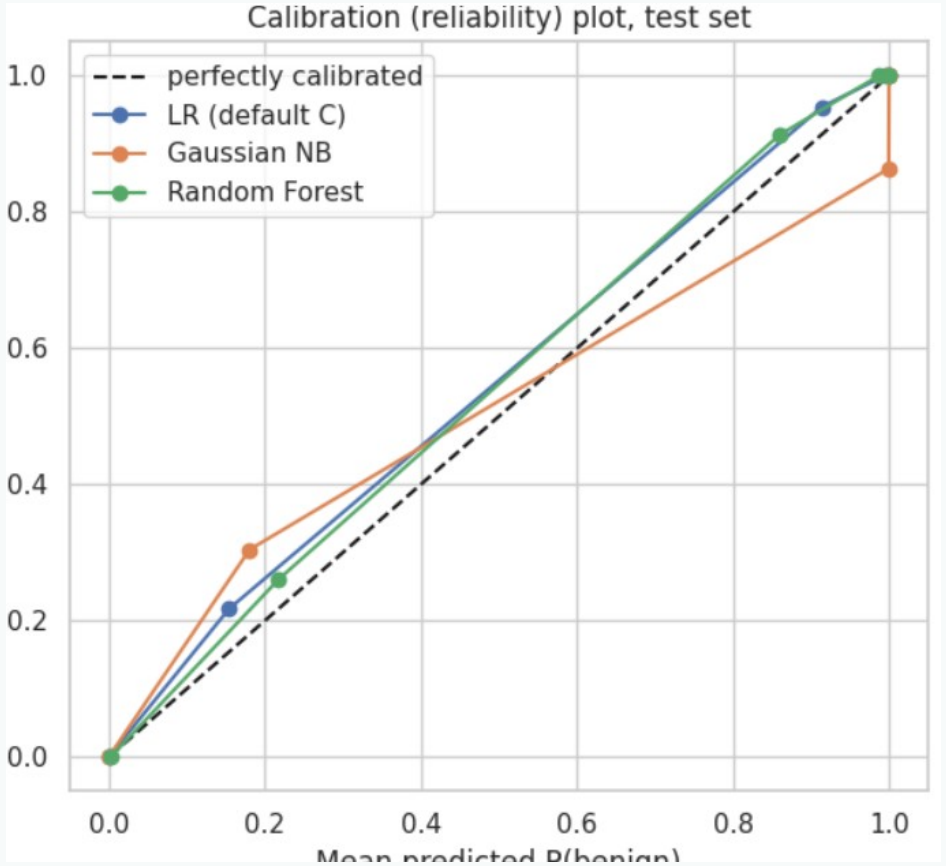


Direction of each feature's effect (beeswarm)

The largest effects came from worst area, worst perimeter and worst concave points. Six of the top 10 important measurements overlapped with Logistic Regression.

Calibration: did predicted probabilities match observed outcomes?

We checked whether predicted probabilities matched actual outcomes. For example, cases assigned about 80% benign probability should be benign about 80% of the time.



Brier score / log-loss (lower is better)

Model	Brier	Log-loss
Logistic Regression	**0.0215**	**0.0777**
Random Forest	0.0319	0.1089
Gaussian NB	0.0602	0.4370

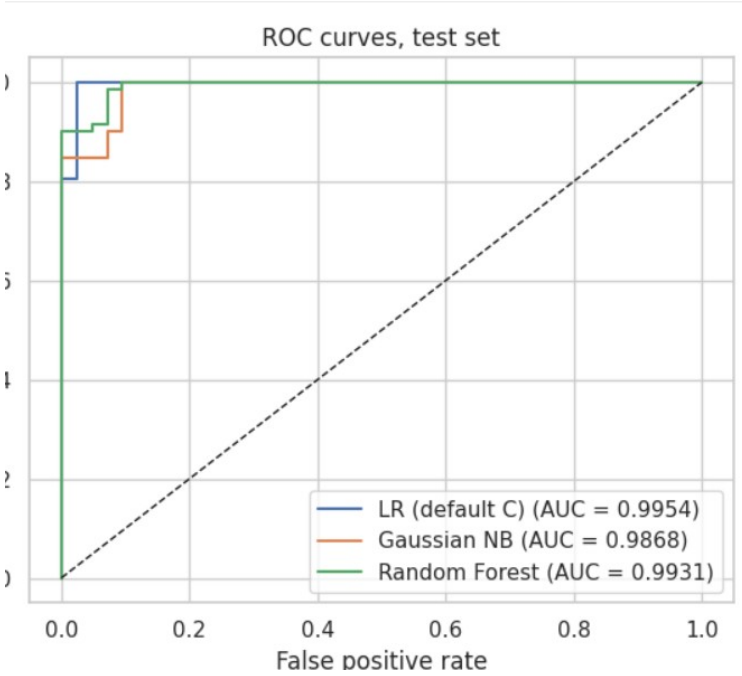
What this tells us

Logistic Regression had the lowest Brier score (0.0215) and log-loss (0.0777), indicating the best probability estimates in this test. The small sample makes calibration curves approximate.

Final comparison of all tested approaches

All four rows use the same 114-patient test set. Accuracy counts correct labels; ROC-AUC measures how well probabilities rank cases; Brier measures probability error (lower is better).

Model	Accuracy	ROC-AUC	Malig. missed	Brier
Logistic Regression	**98.25%**	**0.9954**	**1**	**0.0215**
Random Forest	95.61%	0.9931	3	0.0319
Gaussian Naive Bayes	92.98%	0.9868	4	0.0602
LR, threshold 0.45	95.61%	0.9954	1	0.0215



Changing the threshold affected classifications but not the underlying probabilities, so ROC-AUC and Brier remained unchanged. Accuracy fell, without detecting another malignancy.

Final conclusions and practical limitations

1 Logistic Regression performed best in our comparison

On 114 unseen cases, it achieved 98.25% accuracy and 0.9954 ROC-AUC, with only 1 of the 42 malignant cases missed.

2 Our planned LR–GNB comparison showed a difference

McNemar's exact p-value was 0.0312 for the planned comparison. After correcting for all three pairwise comparisons, the adjusted value was 0.0938.

3 Additional tuning and model complexity did not guarantee gains

Cross-validation selected the original C = 1.0, threshold 0.45 caused more false alarms, and Random Forest performed well but did not beat Logistic Regression.

LIMITATION Single benchmark dataset • 114 test samples (42 malignant) • no external validation • not a medical diagnostic tool